

temp - time dependence:

- Grand unification $\sim 10^{15}$ GeV
- Electroweak unification ~ 250 GeV
- Quark-Hadron transition ~ 1.2 GeV

* Radiation dominant: early times

$$R(t) \sim t^{1/2}$$

→ Can ignore effects of curvature

$$\Omega_{\text{curv.}} \sim \frac{1}{R^2}$$

$$\Omega_r \sim \frac{1}{R^4}$$

$$H(t) = \frac{1}{2t} \quad (\text{early times})$$

• $K=0$: $H^2 = \frac{8\pi G}{3} \rho$

$$2 \sqrt{\frac{8\pi G}{3} \rho} = \frac{1}{t}$$

$$t = \sqrt{\frac{3}{32\pi G}}$$

Bosons: n / $\mathcal{U} = \rho c^2 = \frac{g}{(2\pi\hbar)^3} \int_0^\infty \frac{4\pi p^2 E(p) dp}{e^{G(p)/kT} - 1}$

→ Relativistic: $E(p) \approx pc$

$$\mathcal{U} = \frac{4\pi g c}{(2\pi\hbar)^3} \int_0^\infty \frac{p^3 dp}{e^{pc/kT} - 1}$$

$$y = \frac{pc}{kT}$$

$$\mathcal{U} = \frac{g \pi^2}{30 \hbar^3 c^3} (kT)^4$$

$g \equiv$ effective n of rel. bos. D.O.F.

$$\mathcal{U}_{\text{tot}} = \rho c^2 = \frac{g_* \pi^2}{30 \hbar^3 c^3} (kT)^4$$

$$4\pi g c \int_0^\infty \frac{p^3 dp}{e^{pc/kT} - 1}$$

$$* \mathcal{U}_F = \frac{4\pi}{(2\pi\hbar)^3} \int_0^\infty p^3 f_F(p) dp$$

$$\mathcal{U}_B = \frac{4\pi g_B c}{(2\pi\hbar)^3} \int_0^\infty p^3 f_B(p) dp$$

$$\bullet \frac{1}{e^x + 1} = \frac{1}{e^x - 1} - \frac{2}{e^{2x} - 1}$$

$$\bullet f_F(T) = f_B(T) - 2f_B(T/2)$$

$$\begin{aligned} \frac{\mathcal{U}_F}{g_F} &= \frac{4\pi c}{(2\pi\hbar)^3} \int_0^\infty p^3 (f_B(T) - 2f_B(T/2)) dp \\ &= \frac{1}{g_B} (\mathcal{U}_B(T) - 2\mathcal{U}_B(T/2)) \end{aligned}$$

$$\mathcal{U}_B(T) \propto T^4$$

$$\mathcal{U}_B(T/2) \propto \frac{1}{16} T^4$$

$$\frac{\mathcal{U}_F}{g_F} = \frac{1}{g_B} \left(\mathcal{U}_B(T) - \frac{1}{8} \mathcal{U}_B(T) \right)$$

$$\frac{\mathcal{U}_F}{g_F} = \frac{7}{8g_B} \mathcal{U}_B(T)$$

$$g_F = \sum_B g_{iB} + \frac{7}{8} \sum_F g_{iF}$$

$$* n_F(T) = \frac{3}{4} \left(\frac{g_F}{g_B} \right) n_B(T)$$

$$S_F(T) = \left(\frac{4}{3} \right) \frac{\mathcal{U}_F}{T} \quad \Bigg| \quad S_B(T) = \left(\frac{4}{3} \right) \frac{\mathcal{U}_B}{T}$$

$$\frac{S_F}{S_B} = \frac{\mathcal{U}_F}{\mathcal{U}_B} = \frac{7}{8} \left(\frac{g_F}{g_B} \right)$$

$$S_F = \frac{7}{8} \left(\frac{g_F}{g_B} \right) S_B$$

